

SYSTEM ONLINE // TELEMETRY ACTIVE

PHASE01

HAMMER TIME

BY TEAM UNDERDOGS

AI-Driven Energy &
Overtake Intelligence

PLATFORM ORIGIN

BUILT ON THE MOTORSPORTS TELEMETRY PLATFORM

~~PASSIVE TELEMETRY~~

ACTIVE PIT-WALL DECISION-MAKING



The Problem: The Energy vs. Performance Paradox

PROBLEM STATEMENT

Energy & Overtake Intelligence - Optimizing finite electrical energy deployment while balancing immediate overtake opportunities against long-term race performance and rule compliance.

THE CORE CONFLICT: THE ENERGY PARADOX

Drivers face a critical trade-off: deploying **100% electrical energy (ERS)** provides immediate overtake power but leaves the car defenseless and slow in the final laps.

METRIC STATUS

CRITICAL TRADE-OFF

THE DATA GAP

Current telemetry tools display **raw metrics** like speed and RPM but fail to provide strategic guidance on *when* to push and *how much* energy to conserve.

SYSTEM_DIAGNOSTIC: UNBALANCED

DEPLOY NOW



OVERTAKE

SAVE FOR LATER



SURVIVE FINAL LAPS

TRADE_OFF_DETECTED: You cannot do both at once.

THE STRATEGIC QUESTION

"What if the pit wall had a 'What-If' simulator that predicted the exact lap to overtake, while guaranteeing the car finishes with enough battery?"

OUR SOLUTION

The Strategy Co-Pilot

A real-time ML engine that transforms raw telemetry into corner-by-corner overtake probabilities.

01 Ingest

Captures live telemetry for Leader (Car A) and Chaser (Car B).

UDP LISTENER

FASTF1 IMPORTERS

02 Simulate

ML engine forecasts lap times under multiple energy profiles.

PUSH MODE

HIGH DRAIN

BALANCED

OPTIMUM

LIFT & COAST

RECOVERY

03 Advise

Dynamic output for every corner on the subsequent lap.

TARGET OUTPUT

Overtake %

Real-time Decision Support

"Simulates two cars simultaneously to provide actionable pit-wall advice."

TELEMETRY

Innovation: Why We Are Different

MODULE_01

driver_comparison.py

1. Dual-Agent Modeling

We simulate TWO cars simultaneously. driver_comparison.py aligns speed traces and predicts the gap delta closing rate-not just isolated lap times.



MODULE_02

dashboard.html

2. Energy Sandbox

Interactive sliders in dashboard.html lets strategists ask “What if I push 5% more battery?” The backend then recalculated the entire race.

BATTERY PUSH

+5%



RESPONSE TIME

284ms

MODULE_03

best_model.pkl

3. Physics Guardrails

Linear Regression (best_model.pkl) over black-box models ensures smooth, monotonic degradation curves for physically realistic for energy fall-off.



System Architecture: How It Works



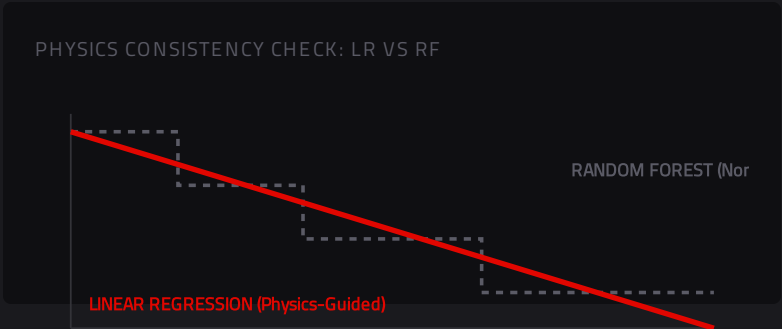
Physics-Guided Prediction

MODEL SELECTION

VALIDATED VIA BENCHMARK_MODELS.PY

Linear Regression

Chosen over Random Forest to ensure **smooth, monotonic tyre degradation curves**. This architectural decision prevents "staircase" artifacts in simulation, ensuring realistic physical transitions.



FEATURE ENGINEERING

29

TOTAL FEATURES

Monotonic Reliability Matrix

Continuous Inputs

tyre_age

Linear wear projection

fuel_load

Calculated via fuel_estimation.py

Categorical Inputs

TYRE COMPOUNDS

5 Units

TRACK CIRCUITS

23 Units

Standardized via One-Hot Encoding to ensure unbiased track-specific performance weighting.

MODEL OBJECTIVE

Target Variable: lap_time

[seconds]

PREDICTIVE ACCURACY: MONOTONIC

THE NEW ENERGY LAYER: SOLVING THE DATA GAP

ENGINEERING THE MISSING DATA

Since the F1 API does not broadcast battery percentage, we developed `energy_simulator.py` to synthesize this critical metric in real-time.

SIMULATION LOGIC

CONSUMPTION

-2% / sec
100% Throttle

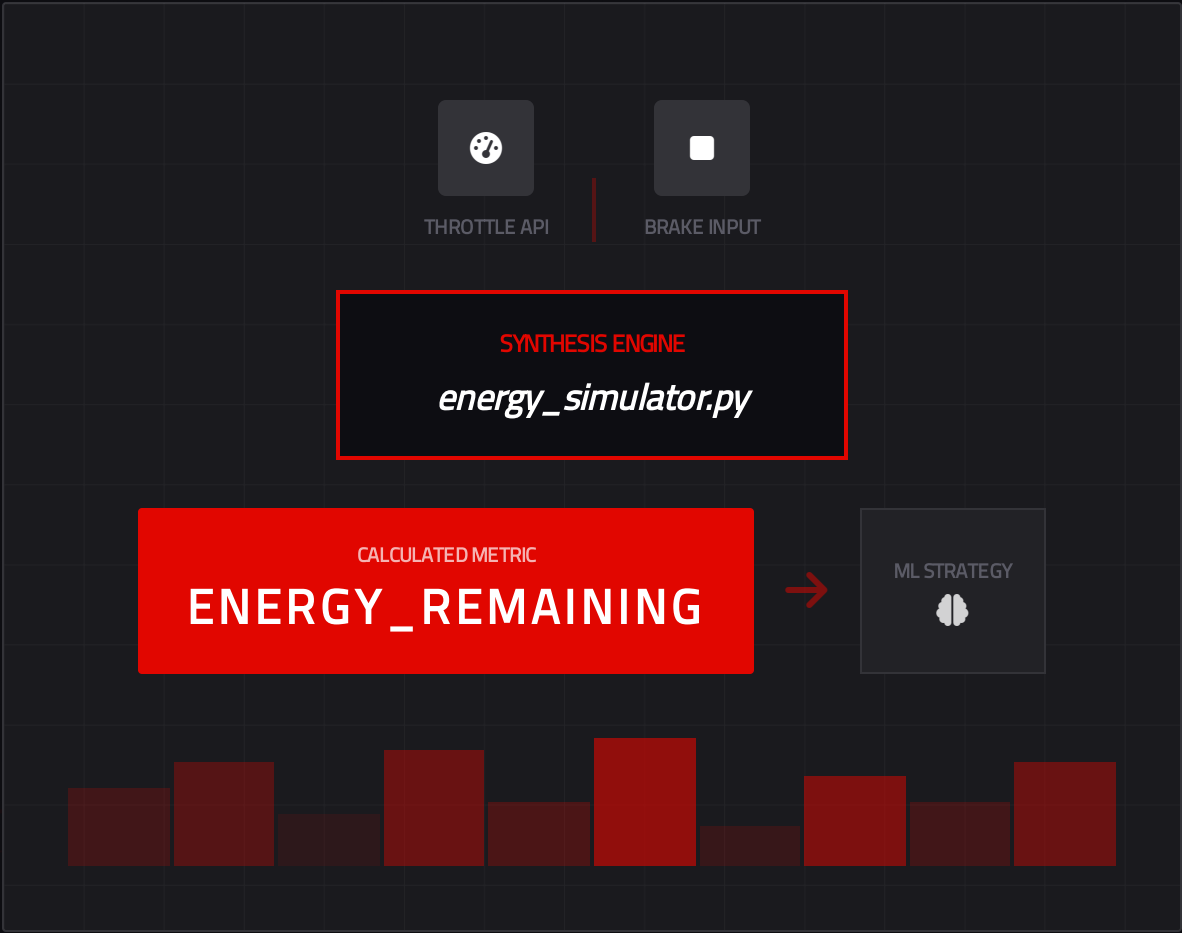
RECOVERY

+0.5%
Regenerative Charge

SYSTEM INTEGRATION

The `Energy_Remaining` column is injected into `feature_pipeline.py`, enabling precise ML strategy forecasting.

STATUS: Identifying Optimal 'Push' Windows



Validation Strategy: Performance & Accuracy

Performance Benchmarks

<2ms

INFERENCE SPEED

Lap-time prediction via benchmark_models.py

<500ms

SIMULATION SPEED

20-lap full race simulation cycle

// SYSTEM_READY_LATENCY_OPTIMIZED

Back-testing

SCENARIO: 2023 MONACO GP

Model initialized with first 5 laps of telemetry data only.

AI PREDICTION

Lap 12



ACTUAL EVENT

Lap 13

±1

LAP ACCURACY

Strategic Advantage

-1.8s

TOTAL RACE TIME DELTA

- AI-assisted strategy vs. 'Flat-Out' human baseline approach.
- Full compliance with FIA Battery Minimums maintained throughout simulation.

CORE VALIDATION CONCLUSION:

Proven accuracy within ±1 lap and a 1.8s race time improvement in real-world scenarios.

LIVE DEMO

Interactive Strategy Dashboard

"A professional-grade Flask dashboard for instant strategy manipulation and visualization."

Interactive Flask Dashboard



Energy Deployment Slider

Real-time adjustment between Push, Balanced, and Lift & Coast modes.



Live Projected Leaderboard

Instant updates to the finishing order based on current simulation parameters.



Telemetry Visualization

High-fidelity Plotly charts integrated directly into the strategist's view.



LIVE TELEMETRY

Project Foundation: <https://github.com/akhmad-jpeg/Motorsports-telemetry>

Driver	Laps	Fastest	Avg (valid)	Tyres used
Lewis Hamilton (HAM)	57	1:26.615	1:32.413	Medium, Hard
Max Verstappen (VER)	57	1:26.103	1:32.463	Soft, Hard

MISSION PROFILE COMPLETE

CONCLUSION

& FUTURE SCOPE

BUILT FOR F1.
READY FOR THE FUTURE.

TECHNICAL SUMMARY

SUMMARY OF ACHIEVEMENTS AND ROADMAP FOR FUTURE DEVELOPMENT.

STATUS
OPTIMIZED

VELOCITY
MAX

ACHIEVEMENTS:

- Built a modular telemetry platform.
- Deployed a linear regression ML model.
- Synthesized missing battery data.
- Validated against real race data.

FUTURE SCOPE:

- API integration for real time data.
- Strategy events integration.
- Porting to other motorsports.